

Pseudo Code:

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D: > For Linux > Academic > 3rd Year 1st Semester > z Our sem > zSessional > Operating Systems Laboratory > Lab4 > SJF.cpp >
1  #include <bits/stdc++.h>
2  using namespace std;
3
4  int main()
5  {
6      int n;
7      cout << "Enter Number of Processes: ";
8      cin >> n;
9
10     vector<int> pid(n), arrival(n), burst(n), completion(n), turnaround(n), waiting(n);
11
12     for (int i = 0; i < n; i++)
13     {
14         pid[i] = i + 1;
15         cout << "Arrival Time of Process " << i + 1 << ": ";
16         cin >> arrival[i];
17         cout << "Burst Time of Process " << i + 1 << ": ";
18         cin >> burst[i];
19     }
20
21     // Sort processes
22     vector<int> idx(n);
23     for (int i = 0; i < idx.size(); i++)
24     {
25         idx[i] = i;
26     }
27     // sort(idx.begin(), idx.end(), [&](int a, int b)
28     //     { return arrival[a] < arrival[b]; });
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29     for (int i = 0; i < idx.size() - 1; i++)
30     {
31         int minPos = i;
32         for (int j = i + 1; j < idx.size(); j++)
33         {
34             if (arrival[idx[j]] < arrival[idx[minPos]])
35             {
36                 minPos = j;
37             }
38         }
39
40         // swap
41         int temp = idx[i];
42         idx[i] = idx[minPos];
43         idx[minPos] = temp;
44     }
45
46     vector<int> sortedPid(n), sortedArrival(n), sortedBurst(n);
47     for (int i = 0; i < n; i++)
48     {
49         sortedPid[i] = pid[idx[i]];
50         sortedArrival[i] = arrival[idx[i]];
51         sortedBurst[i] = burst[idx[i]];
52     }
53
54     vector<int> grantChart;
55     int currentTime = 0;
56     int idleTime = 0;
57     int completed = 0;
58     vector<bool> isCompleted(n, false);
59
```

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60 while (completed < n)
61 {
62     vector<pair<int, int>> readyQueue;
63     for (int i = 0; i < n; i++)
64     {
65         if (sortedArrival[i] <= currentTime && !isCompleted[i])
66         {
67             readyQueue.push_back({sortedBurst[i], i});
68         }
69     }
70     sort(readyQueue.begin(), readyQueue.end());
71     if (readyQueue.empty())
72     {
73         // idle detect
74         int nextArrival = INT_MAX;
75         for (int i = 0; i < n; i++)
76         {
77             if (!isCompleted[i] && sortedArrival[i] > currentTime)
78             {
79                 nextArrival = min(nextArrival, sortedArrival[i]);
80             }
81         }
82         idleTime += (nextArrival - currentTime);
83         cout << endl;
84         cout << "CPU Idle from " << currentTime << " to " << nextArrival << endl;
85         currentTime = nextArrival;
86         continue;
87     }
88
89     int idx2 = readyQueue[0].second;
90
91     grantChart.push_back(sortedPid[idx2]);
92     completion[idx2] = currentTime + sortedBurst[idx2];
93     turnaround[idx2] = completion[idx2] - sortedArrival[idx2];

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93     turnaround[idx2] = completion[idx2] - sortedArrival[idx2];
94     waiting[idx2] = turnaround[idx2] - sortedBurst[idx2];
95     currentTime = completion[idx2];
96     isCompleted[idx2] = true;
97     completed++;
98 }
99 // Output Results
100 cout << endl;
101 cout << "Grant Chart" << endl;
102 int abrar = grantChart.size();
103 for (int i = 0; i < abrar; i++)
104 {
105     cout << "P" << grantChart[i] << "|";
106 }
107 cout << endl;
108 cout << endl;
109 cout << "Total Idle Time: " << idleTime << " units" << endl;
110 float avgTurnaround = 0, avgWaiting = 0;
111
112 for (int i = 0; i < n; i++)
113 {
114     cout << "Process " << sortedPid[i] << ": Arrival Time = " << sortedArrival[i] << ", Burst Time = " << sortedBurst[i] << ", Completion Time = " << completion[i]
115     avgTurnaround += turnaround[i];
116     avgWaiting += waiting[i];
117 }
118 avgTurnaround /= n;
119 avgWaiting /= n;
120 cout << "\nAverage Turnaround Time: " << avgTurnaround << endl;
121 cout << "Average Waiting Time: " << avgWaiting << endl;
122
123 return 0;
124 }

```

Input/Output:

```
C:\WINDOWS\system32\cmd.  X  +  v

Enter Number of Processes: 4
Arrival Time of Process 1: 0
Burst Time of Process 1: 6
Arrival Time of Process 2: 1
Burst Time of Process 2: 8
Arrival Time of Process 3: 2
Burst Time of Process 3: 7
Arrival Time of Process 4: 3
Burst Time of Process 4: 3

Grant Chart
P1||P4||P3||P2||

Total Idle Time: 0 units
Process 1: Arrival Time = 0, Brust Time = 6, Completion Time = 6, Turnaround Time = 6, Waiting Time = 0
Process 2: Arrival Time = 1, Brust Time = 8, Completion Time = 24, Turnaround Time = 23, Waiting Time = 15
Process 3: Arrival Time = 2, Brust Time = 7, Completion Time = 16, Turnaround Time = 14, Waiting Time = 7
Process 4: Arrival Time = 3, Brust Time = 3, Completion Time = 9, Turnaround Time = 6, Waiting Time = 3

Average Turnaround Time: 12.25
Average Waiting Time: 6.25

Press any key to continue . . .
```